GND
VCC

Pin header connections

J1 EPSM_ADDR

GND	1	2	VCC
B_CPU_A11	3	4	B_M2
B_CPU_A10	5	6	B_CPU_A12
B_CPU_A9	7	8	B_CPU_A13
B_CPU_A8	9	10	B_CPU_A14
B_CPU_A7	11	12	
B_CPU_A6	13	14	EPSM_A1
B_CPU_A5	15	16	EPSM_A0
B_CPU_A4	17	18	EPSM_CE3
B_CPU_A3	19	20	EPSM_CE2
B_CPU_A2	21	22	EPSM_CE1
B_CPU_A1	23	24	
B_CPU_A0	25	26	GND
B_CPU_RW	27	28	B_ROMSEL
	29	30	GND

J2 EPSM_DATA

GND	1	2	VCC
YM_CS	3	4	B_CPU_D7
YM_A0	5	6	B_CPU_D6
YM_A1	7	8	B_CPU_D5
YM_D7	9	10	B_CPU_D4
YM_D6	11	12	B_CPU_D3
YM_D5	13	14	B_CPU_D2
YM_D4	15	16	B_CPU_D1
YM_D3	17	18	B_CPU_D0
YM_D2	19	20	EPSM_A1
YM_D1	21	22	EPSM_A0
YM_D0	23	24	EPSM_CE3
GND	25	26	EPSM_CE2
GND	27	28	EPSM_CE1
VCC	29	30	GND

Defaults to $4016/$401C-F addr
To use different addressing, disconnect jumpers.

JP1	GND	EPSM_CE1
JP2	$401CFWCE	EPSM_CE2
JP3	VCC	EPSM_CE3
JP4	B_CPU_A2	EPSM_A0
JP5	B_CPU_A1	EPSM_A1

Famicom bus lines must be buffered to 5V CMOS!

$4016/$401C-F multiplexer

U9 74HC157
U13 74HC157
U14 74HC157

	U9	U13	U14
I0a	EPSM_CS	B_CPU_D6	B_CPU_D2
I1a	S_CS	S_CPU_D6	S_CPU_D2
I0b	EPSM_A1	B_CPU_D4	B_CPU_D0
I1b	S_A1	S_CPU_D4	S_CPU_D0
I0c	B_CPU_D7	B_CPU_D3	GND
I1c	S_CPU_D7	S_CPU_D3	VCC
I0d	EPSM_A0	B_CPU_D5	B_CPU_D1
I1d	S_A0	S_CPU_D5	S_CPU_D1
S	$4016_P	$4016_P	$4016_P
E	GND	GND	GND
Za	4 YM_CS	4 YM_D6	4 YM_D2
Zb	7 YM_A1	7 YM_D4	7 YM_D0
Zc	9 YM_D7	9 YM_D3	9
Zd	12 YM_A0	12 YM_D5	12 YM_D1

R4 470
D1 PINK
C12 0.1u
C15 0.1u
C17 0.1u

$4016 pulse delay control select

U10A 74HC132
U10B 74HC132
U10C 74HC132
U10D 74HC132
U10E 74HC132

$4016_P — R1 3K — C9 68p — GND
$4016_P — R2 1K — C10 68p — GND
$4016_P_DL — R3 1K — C11 68p — GND

U10A: 1 $4016_P, 2 C9, 3 $4016_P_DL
U10B: 4 $4016_P, 5 $4016_P_DL, 6 S_CS
U10D: 12, 13, 11
U10C: 9, 10, 8 S_RST
U10E: 14 VCC, 7 GND

C16 0.1u

74'138 address decoding

U1 74HC138
U5 74HC138
U2 74HC138
U6 74HC138

U1: B_CPU_A14, B_CPU_A12, B_CPU_A11, B_CPU_A13, B_CPU_RW, B_M2 → $40xx_W_A
U5: B_CPU_A6, B_CPU_A7, B_CPU_A8, B_CPU_A9, B_CPU_A10, B_ROMSEL → $40xx_W_B
U2: B_CPU_A0, B_CPU_A1, B_CPU_A2, B_CPU_A3, B_CPU_A5, B_CPU_A4 → $4016_CE
U6: B_CPU_A2, B_CPU_A3, GND, GND, B_CPU_A5, B_CPU_A4 → $401C-F_CE

C1 0.1uF
C5 0.1uF
C2 0.1uF
C6 0.1uF

$40xx_W_A
CPU_A14 = 1
CPU_A12 = 0
CPU_A11 = 0
CPU_A13 = 0
CPU_RW = 0
M2 = 1

$40xx_W_B
CPU_A6 = 0
CPU_A7 = 0
CPU_A8 = 0
CPU_A9 = 0
CPU_A10 = 0
ROMSEL = 1

$4016_W
CPU_A1 = 1
CPU_A2 = 1
CPU_A0 = 0
CPU_A3 = 0
CPU_A5 = 0
CPU_A4 = 1

$401C-F_W
CPU_A2 = 1
CPU_A3 = 1
CPU_A5 = 0
CPU_A4 = 1

U8A 74HC32: $40xx_W_A, $40xx_W_B → $40xx_W_CE
U8B 74HC32: $40xx_W_CE, $401C-F_CE → $401CFWCE
U8D 74HC32: $40xx_W_CE, $4016_CE → $4016_W_CE
U8C 74HC32: GND, GND → 8
U8E 74HC32: VCC, GND, VCC
C8 0.1uF

U11 74LVC1GU04: $4016_W_CE → 4 $4016_W_CE
C13 0.1u

<3)~

Alternative 74'688 address decoding

U15 74HC688
U16 74HC688
U17 74HC688

U15: B_CPU_A9, B_CPU_A12, B_CPU_A10, B_CPU_A11, B_CPU_A8, B_CPU_A14, B_CPU_A13, B_ROMSEL; P=R $40xx_CE
U16: B_CPU_A2, B_CPU_A3, B_CPU_A4, B_CPU_A5, B_CPU_A6, B_CPU_A7, B_CPU_RW, B_M2; P=R $401CFWCE
U17: B_CPU_A0, B_CPU_A1, B_CPU_A2, B_CPU_A3, B_CPU_A4, B_CPU_A5, B_CPU_A6, B_CPU_A7; P=R $4016_CE

$40xx
CPU_A8 = 0
CPU_A9 = 0
CPU_A10 = 0
CPU_A11 = 0
CPU_A12 = 0
CPU_A13 = 0
CPU_A14 = 1
ROMSEL = 1

$401C-F W
CPU_A2 = 1
CPU_A3 = 1
CPU_A4 = 1
CPU_A5 = 0
CPU_A6 = 0
CPU_A7 = 0
CPU_RW = 0
M2 = 1

$4016
CPU_A0 = 0
CPU_A1 = 1
CPU_A2 = 1
CPU_A3 = 0
CPU_A4 = 1
CPU_A5 = 0
CPU_A6 = 0
CPU_A7 = 0

C18 0.1u
C19 0.1u
C20 0.1u

U18 74HCT1G02: $4016_CE, B_CPU_RW → $4016_OUT1
U19 74HCT1G08: $4016_OUT1, B_M2 → $4016_W_CE
C21 0.1u
C22 0.1u

$4016 OUT1 data/address latch

U3 74HC174
U7 74HC174

U3: B_CPU_D7, B_CPU_D6, B_CPU_D5, B_CPU_D4, B_CPU_D3, B_CPU_D2 → S_CPU_D7, S_CPU_D6, S_CPU_D5, S_CPU_D4, S_A0, S_A1
U7: VCC, B_CPU_D7, B_CPU_D6, B_CPU_D5, B_CPU_D4, VCC → $4016_P, S_CPU_D3, S_CPU_D2, S_CPU_D1, S_CPU_D0

OUT1 → Cp
S_RST → Mr

C3 0.1u
C7 0.1u

U4A 74HC139
U4B 74HC139
U4C 74HC139

U4A: OUT1, GND → OUT1
U4B: EPSM_CE3, EPSM_CE2, EPSM_CE1 → EPSM_CS
C4 0.1u

OUT1 latch reconstruction

U12 74LVC1G373

B_CPU_D1 → D
$4016_W_CE1 → LE
OUT1 ← Q
OE → GND

C14 0.1u

Some pinouts subject to change over time for ease of layout
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Title: EPSM-FC $4016/$401C-F Address Decoder Daughterboard

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